

LINEAR ALGEBRA

Foundations for Applied Statistics & Machine Learning

Theory, Formulas & Visual Concepts

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Project Learning Outcomes

By completing this project, students will achieve the following:

- Understand vector and matrix manipulation using real-world data.
- Apply norms, dot/cross products, and projections to performance data.
- Gain insight into the meaning of eigenvalues and eigenvectors.
- Simplify and interpret data by implementing LU, SVD, PCA, and LDA.
- Build confidence in using linear algebra foundations for applied statistics and machine learning.

1. Vectors and Matrices

1.1 What Is a Vector

Definition

A vector is a quantity that has both magnitude (size) and direction. In data science, a data point is represented as a vector — for example, a customer's 'age' and 'income' features can be shown as the vector [age, income].

$$\mathbf{v} = [v_1, v_2, \dots, v_n] \in \mathbb{R}^n$$

1.2 What Is a Matrix

Definition

A matrix is a rectangular arrangement of numbers organized into rows and columns. It lets us represent multiple data points (rows) together with their features (columns) at once — an entire dataset is essentially a matrix.

$$\mathbf{A} (m \times n) = [a_{ij}], i = 1 \dots m, j = 1 \dots n$$

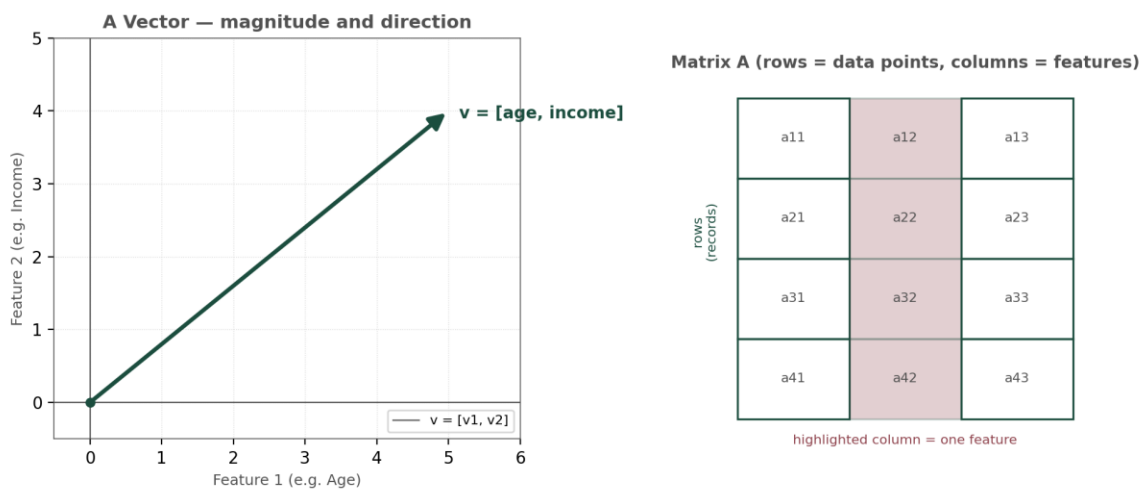


Figure 1.1 — A vector shown as magnitude and direction, and a matrix as rows (records) and columns (features)

1.3 Basic Matrix Operations

- Addition: $(\mathbf{A} + \mathbf{B})_{ij} = A_{ij} + B_{ij}$ — matrices of the same shape are added element-wise.
- Scalar Multiplication: $(c\mathbf{A})_{ij} = c \cdot A_{ij}$
- Matrix Multiplication: $(\mathbf{AB})_{ij} = \text{Sum over } k \text{ of } A_{ik} \cdot B_{kj}$ — the number of columns in A must equal the number of rows in B.
- Transpose: $(\mathbf{A}^T)_{ij} = A_{ji}$ — rows and columns are interchanged.

Basic Matrix Operations



Figure 1.2 — The four basic matrix operations: addition, scalar multiplication, multiplication, and transpose

2. Norms

Definition

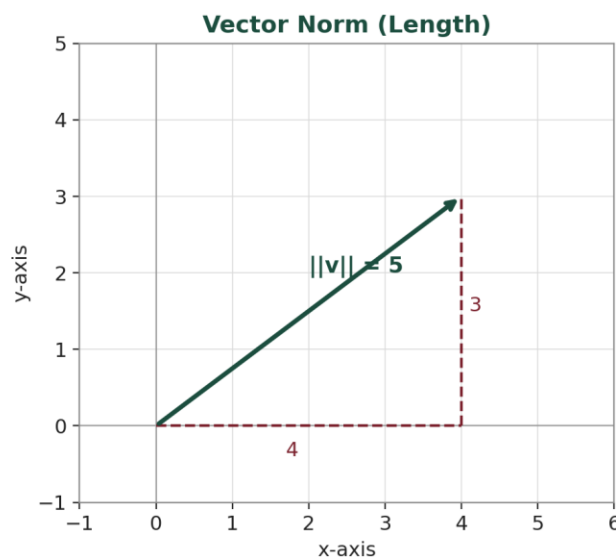
A norm measures the 'length' or 'magnitude' of a vector. In data science, it is used to calculate distance, error, and regularization (as in Ridge/Lasso regression).

2.1 Common Norms

L1 Norm: $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$

L2 Norm (Euclidean): $\|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$

L ∞ Norm: $\|v\|_\infty = \max(|v_1|, |v_2|, \dots, |v_n|)$

Figure 2.1 — Vector $v = (4, 3)$, $\|v\|_2 = 5$ (from the Pythagorean theorem)

3. Dot Product and Cross Product

3.1 Dot Product

Definition

The dot product multiplies two vectors together to give a single scalar number. It shows how much two vectors point in a 'similar direction' — widely used in cosine similarity, correlation, and neural network layers.

$$a \cdot b = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$a \cdot b = \|a\| \|b\| \cos(\theta)$$

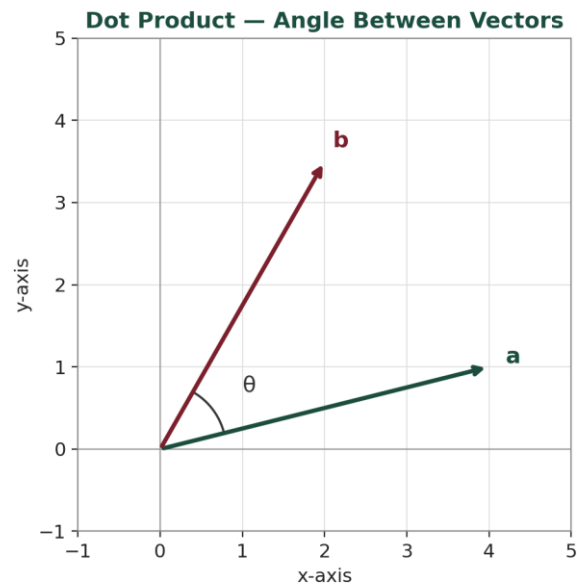


Figure 3.1 — The angle θ between two vectors, derived from the dot product

3.2 Cross Product

Definition

The cross product is defined only for 3D vectors, and it produces a new vector that is perpendicular to both original vectors. Its magnitude gives the area of the parallelogram formed by those two vectors.

$$a \times b = \|a\| \|b\| \sin(\theta) \cdot \hat{n}$$

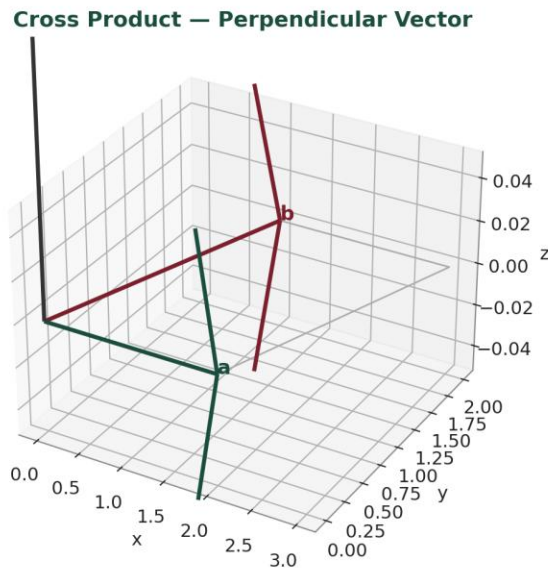


Figure 3.2 — $a \times b$, in the direction perpendicular to both vectors

4. Vector Projections

Definition

A projection shows how much one vector 'extends' in the direction of another vector. This is the base concept behind regression, dimensionality reduction (PCA), and feature transformation.

$$\text{proj}_a(b) = ((a \cdot b) / \|a\|^2) \cdot a$$

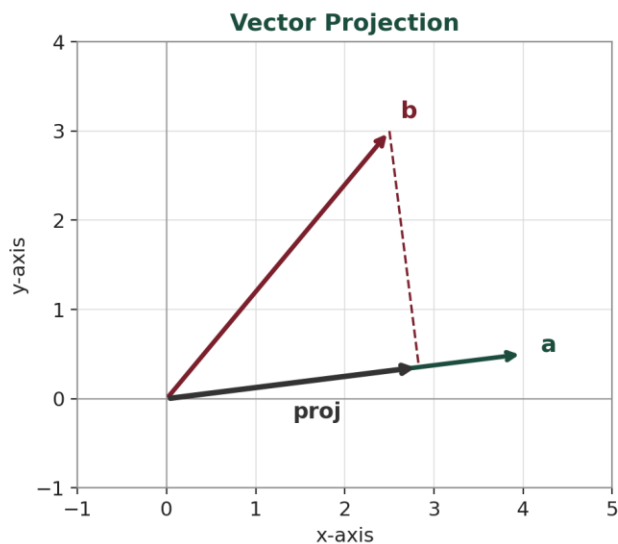


Figure 4.1 — Projection of vector b onto vector a

5. Eigenvalues and Eigenvectors

Definition

An eigenvector is a special vector whose direction does not change even after a matrix transformation — only its length is scaled, by a factor known as the eigenvalue. Dimensionality reduction techniques such as PCA are based on this concept.

$$A \mathbf{v} = \lambda \mathbf{v}$$

$$\det(A - \lambda I) = 0$$

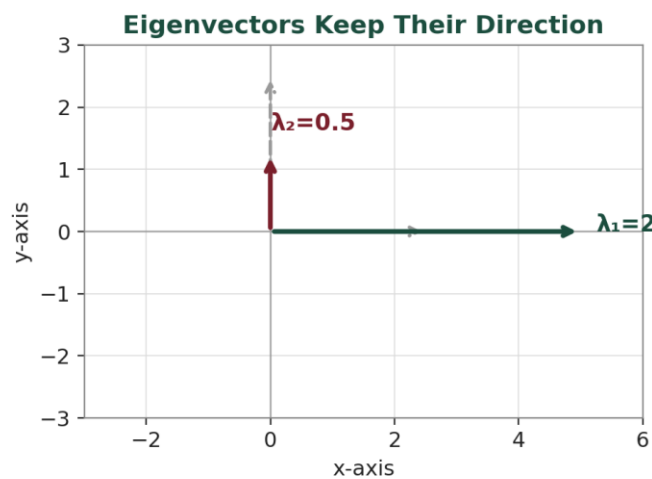


Figure 5.1 — Eigenvectors maintain their direction; they are only scaled by a factor of λ

6. LU Decomposition

Definition

LU Decomposition breaks a matrix A into two simpler matrices: L (lower triangular) and U (upper triangular). This makes solving linear equations, finding the matrix inverse, and calculating the determinant much easier.

$$A = L \cdot U$$

LU Decomposition: $A = L \times U$ Figure 6.1 — Matrix A decomposed into Lower (L) and Upper (U) triangular matrices**7. Singular Value Decomposition (SVD)****Definition**

SVD decomposes any matrix (square or non-square) into three matrices. It is a core technique in dimensionality reduction, image compression, and recommendation systems (such as Netflix and Amazon).

$$A = U \cdot \Sigma \cdot V^T$$

- U — left singular vectors (orthogonal matrix)
- Σ — singular values (diagonal matrix, ordered by importance)
- V^T — right singular vectors (orthogonal matrix)

Singular Value Decomposition: $A = U \Sigma V^T$ Figure 7.1 — Decomposition of matrix A into U , Σ , and V^T **8. Principal Component Analysis (PCA)****Definition**

PCA is a dimensionality reduction technique that projects high-dimensional data onto the 'directions' (principal components) with the most variance. This reduces the complexity of the data without losing much information.

$$\text{Covariance Matrix: } \Sigma = (1/n) X^T X$$

$$\Sigma v_i = \lambda_i v_i \text{ (eigen-decomposition of the covariance matrix)}$$

Steps: (1) Standardize the data → (2) Compute the covariance matrix → (3) Find eigenvalues/eigenvectors → (4) Select the eigenvectors (principal components) with the largest eigenvalues → (5) Project the data onto those components.

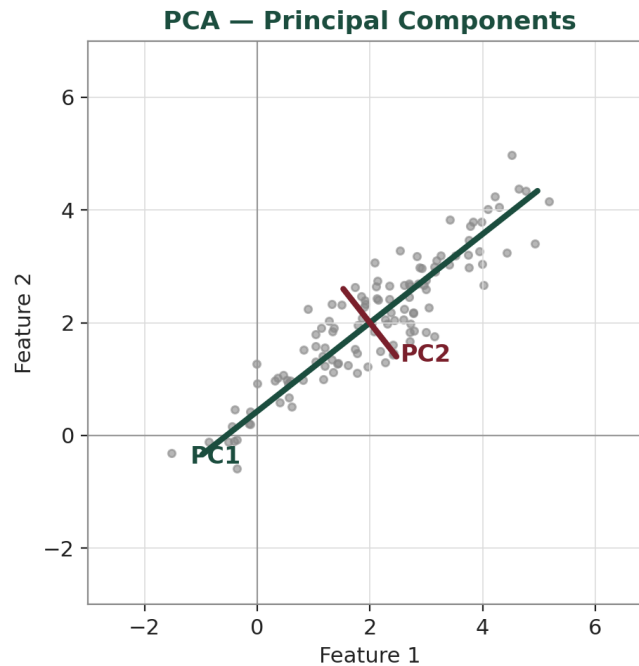


Figure 8.1 — PC1 (highest variance) and PC2 directions

9. Linear Discriminant Analysis (LDA)

Definition

LDA is also a dimensionality reduction technique, but unlike PCA, it maximizes the separation between classes (supervised), whereas PCA only maximizes overall variance (unsupervised). It is widely used in classification problems.

$$J(w) = (w^T S_b w) / (w^T S_w w)$$

S_b = Between-class scatter, S_w = Within-class scatter

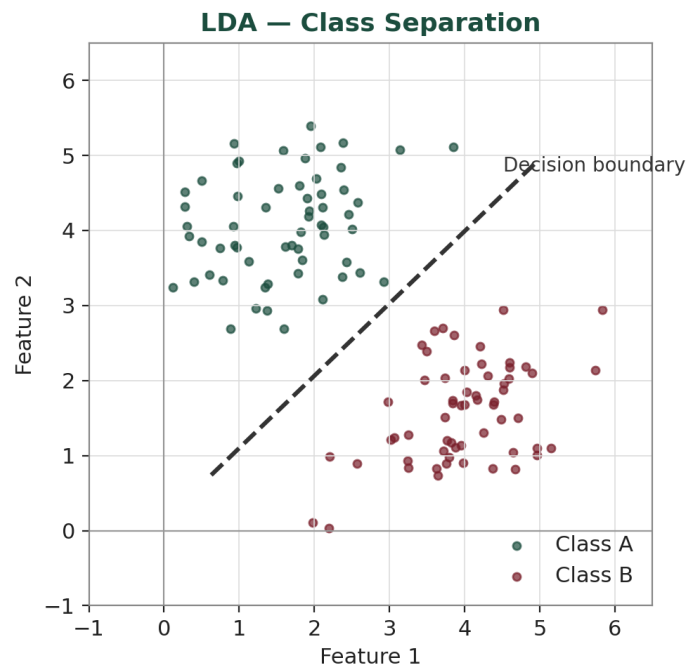


Figure 9.1 — LDA finds the direction that best separates the classes

10. Quick Summary

Concept	Core Formula	Use in Data Science
Vector Norm	$\ v\ _2 = \sqrt{\sum v_i^2}$	Distance, regularization
Dot Product	$a \cdot b = \sum a_i b_i$	Similarity, correlation
Eigenvectors	$Av = \lambda v$	PCA, stability analysis
LU Decomposition	$A = LU$	Solving linear systems
SVD	$A = U \Sigma V^T$	Compression, recommenders
PCA	$\Sigma v = \lambda v$	Dimensionality reduction
LDA	$\max (w^T S_b w) / (w^T S_w w)$	Supervised class separation